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WR418: Final Group Project

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Analysis of the South Platte River Basin

1. Introduction

** Unless otherwise indicated, the information in this introductory section is drawn from the South Platte Basin Context section of the Colorado Water Plan, published by the Colorado Water Conservation Board, 2023. **

The South Platte Basin houses around 70 percent of Colorado's population, with the state's economy driven by urban business and industry mainly in the Front Range of this basin. Three of Colorado's largest cities, Denver, Aurora, and Fort Collins, are in the South Platte Basin. This makes this watershed Colorado's most industrialized basin. The greatest concentration of Colorado's irrigated agricultural land also falls in the South Platte Basin, which contains over 850,000 irrigated acres. The diverse topography of this basin originates in the northern mountainous portion of the Front Range and emerges southwest of Denver's metropolitan area. The main stem goes through the Denver urban area, and moves east across the grassland and cultivated fields of the High Plains. The South Platte River exits Colorado and enters Nebraska near Julesburg merging with the North Platte River. This basin is home to a variety of ecosystems and valuable water-dependent ecological and recreational attributes. Its environmental features provide many recreational opportunities for Colorado's locals and visitors, including major destinations like Rocky Mountain National Park. The Foothills region offers abundant opportunities for Gold Medal fishing, rafting, and skiing. *Figure 1* below from the Colorado Water Science Center shows a map of the South Platte River Basin, its tributary streams, and the land cover and usage in different areas.

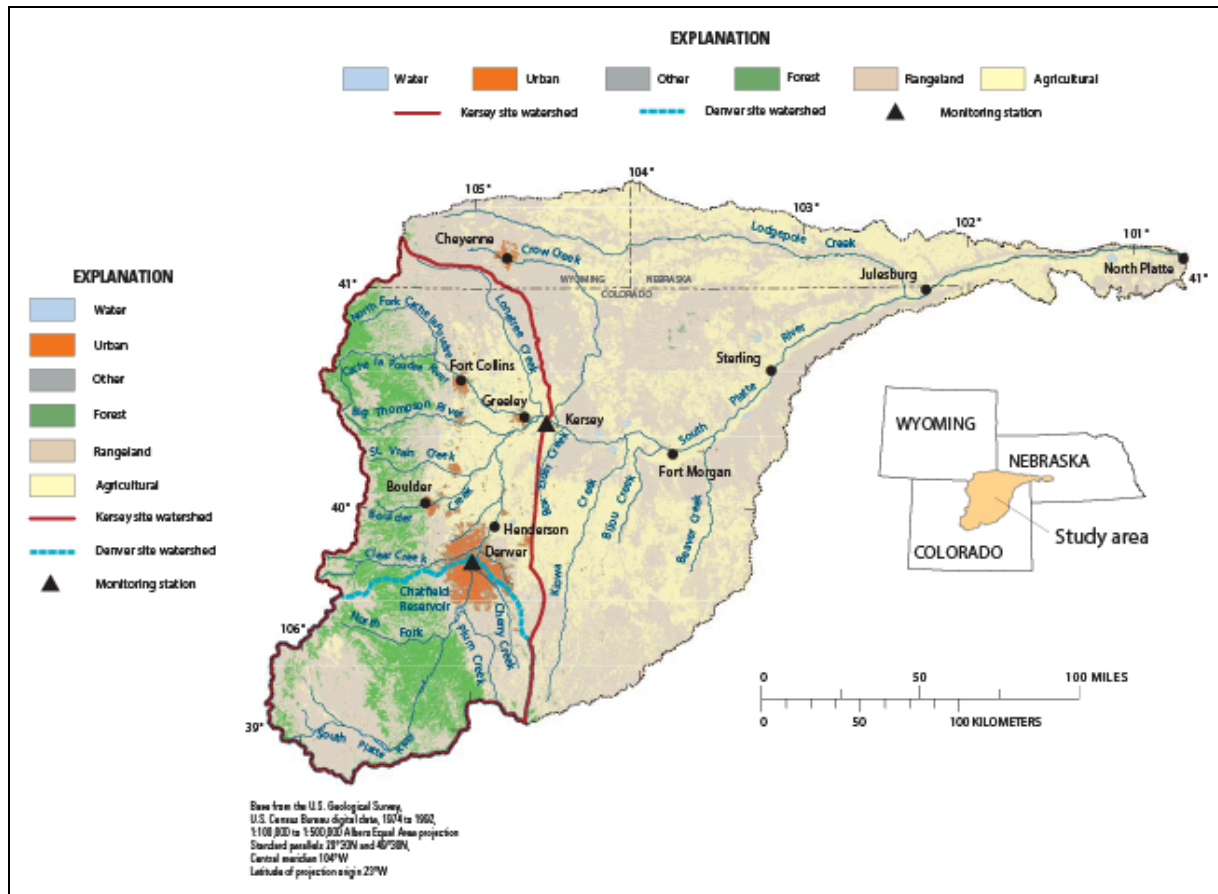


Figure 1: Map showing stream network and land use/cover in the South Platte River Basin. We can see the watershed has forested areas in the mountains, urban areas along the Front Range corridor, and agricultural areas in the plains (Colorado Water Science Center, n.d.).

This basin's precipitation falls as rain during late spring and snow during the winter. In between these periods, the South Platte Basin endures dry conditions. Between October and April, the lower plains receive an average of 3 inches of precipitation, while the mountains average at 22 inches. For May through September, the lower plains typically receive about 6 inches of precipitation, compared to the 15 inches in the mountains. The South Platte Basin's flow changes seasonally; around 70 percent of annual streamflow occurs during spring runoff as the snow melts in the mountains, and the streams have lower flow levels during the remaining parts of the year. The South Platte Basin's groundwater system consists primarily of a shallow unconfined alluvial aquifer along the mainstem and tributaries. There are also deeper, confined Denver Basin aquifers below some areas of the South Platte Basin. The basin's water availability is complicated and greatly influenced by the large population living within this watershed. Davitt's study on climate variability and drought in the South Platte River Basin (2011) revealed "above-average use for public consumption in surface and ground water and above-average agriculture groundwater consumption". Davitt states that in past drought years, water-conserving responses consisted mainly of municipal restrictions, ignoring possible opportunities for improvements in irrigation methods or switching to dry-land crops. This had notable impacts on

the economy, lowering harvests and decreasing revenue in the region. Since this study in 2011, the Basin's water conservation and management methods have improved to better respond to drought and the water conflicts and shortages that come with it.

Water scarcity, rapid growth, and future uncertainty are challenges to the already over-allocated water resources in the South Platte Basin. This has prompted innovative water supply solutions; the basin system's setup demonstrates a very efficient use and reuse of water supplies, successfully taking the basin's hydrology into account. Because more water is diverted from the basin than its native flow, we know that it relies heavily on return flow. Due to rapid growth in Denver's southern metropolitan area, particularly in Douglas and Arapahoe counties, there is an increasing reliance on nonrenewable groundwater. This competition for scarce water supplies is increasing water costs. Meeting the intense municipal, industrial, and agricultural water demands in the basin requires extensive storage systems, groundwater extraction, and transmountain diversions from the West Slope to manage and supplement in-basin water supplies while still maintaining environmental and recreational resources. Meeting these needs can be challenging, possibly leading to conflicts forcing water users to prioritize certain existing and future needs over others. This challenge could worsen with climate change, which introduces significant uncertainty, potentially raises water demand across all sectors, reduces available supply, and presents resilience issues that must be addressed in present-day planning. Additionally, it is important to keep in mind that water supply solutions and strategies may create unintended consequences for other water users in the South Platte Basin.

Water resource management in the South Platte Basin is a complex, multi-purpose effort that balances agricultural, municipal, industrial, environmental, and recreational needs. Historically, water development began in the late 1800s with irrigation infrastructure and was formalized through interstate agreements like the South Platte River Compact (1923) and the Republican River Compact (1943). Over time, major projects such as the Moffat Tunnel, Colorado–Big Thompson Project, and Harold D. Roberts Tunnel were developed to transfer water across basins to meet growing Front Range demands. Current water resource management efforts include modern agricultural upgrades, municipal waterline expansions, raw water development, and stream restoration projects. Collaborative efforts like the Platte River Recovery Implementation Program (PRRIP) help protect threatened and endangered species while also supporting water users across the Basin.

To address the growing water challenges in the region, the South Platte and Metro Basin Roundtables have developed a strategic vision for the future centered on protecting and enhancing all water uses while increasing public awareness. Their goals include closing the municipal supply gap, sustaining irrigated agriculture, and improving watershed health. The Roundtables also emphasize the importance of conservation, reuse, and efficient water management. In addition, they aim to support collaborative efforts to develop water supply options beyond the basin, use scenario planning to navigate future uncertainties, and strengthen outreach, education, and permitting processes to support long-term water planning and implementation.

2. Hypothesis

1. Because the mainstem of the South Platte Basin flows through Denver's urban area, we expect to see changes in the water quality such as an increase in concentrations of pollutants, reduced dissolved oxygen, and alterations in the water's physical properties like turbidity.
2. We suspect that discharge will increase as you leave the headwaters of the South Platte, but as it flows through Denver the amount of acre feet passing should decrease due to consumption. We also suspect the South Platte to have more water than the North Platte due to its length and direction of flow.

3. Data Analysis

Site Selection

We analyzed data collected from USGS monitoring sites along the South Platte River, North Platte River, and Platte River. Parameters analyzed include temperature and discharge data. Temperature data was collected monthly, with skipping winter months as this site only is seasonal. Discharge data was collected every 15 minutes. Site selection included locations with the most data available regarding the parameters of interest. Seven sites in total were selected: Hoosier Pass, Colorado (USGS site 09044800) consisting of 8,105 observations, Strontia Springs, Colorado (USGS site 06707525) with 8,780 observations, Englewood, Colorado (USGS site 06711565) with 13,484 observations, Commerce City, Colorado (USGS site 06714215) consisting of 13,515 observations, Fort Morgan, Colorado (USGS 06759500 site) with 6,600 observations, the Wyoming/Nebraska Border (USGS site 06674500) consisting of 13,514 observations, and Overton, Nebraska (USGS site 0676800) with 13,515 observations (USGS n.d.).

Temperature

To examine the influence of urban development on river conditions, we compared water temperature data from two USGS sites on the South Platte River: Strontia Springs (06707525), representing a relatively undisturbed, forested headwaters region, and Englewood (06711565), situated downstream in the heart of the Denver metropolitan area. Strontia Springs data was collected from April 1, 2022, to October 3, 2024, and Englewood data was collected from February 1, 1983, to October 4, 2024. This analysis focuses on how urbanization impacts physical water properties, specifically temperature, through annual, monthly, and continuous time series comparisons.

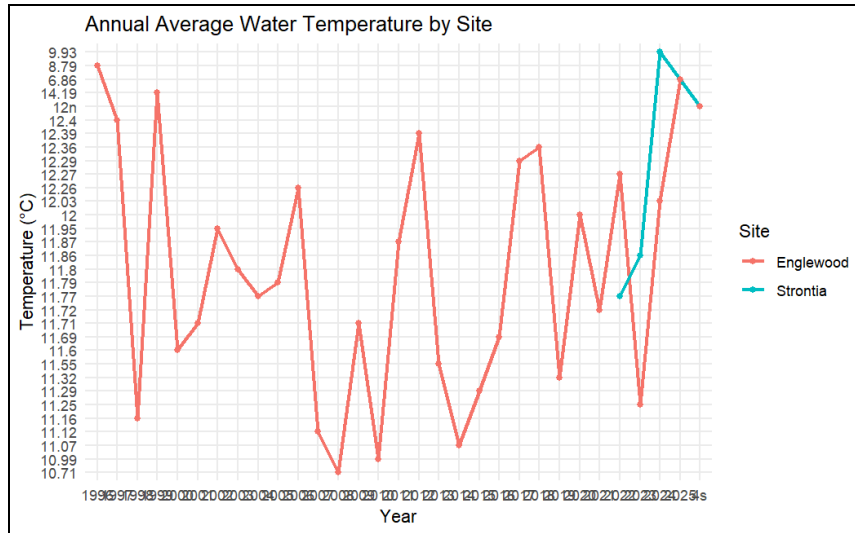


Figure 2: Plot showing annual average water temperature trends of the Englewood and Strontia Springs sites in the South Platte Basin. Englewood data is from 1983-2024 and Strontia Springs data is from 2022-2024.

Our hypothesis suggests that water quality deteriorates as the river flows through urban areas like Denver due to increased tight surfaces, reduced riparian shading, and heat inputs. Annual temperature comparisons support the fact that Englewood consistently shows higher mean yearly temperatures than Strontia Springs. This increase is likely the result of both natural warming along the river's descent and added urban heat. The temperature differential between sites is more pronounced in some years, potentially reflecting years of lower flow or intensified urban heat effects.

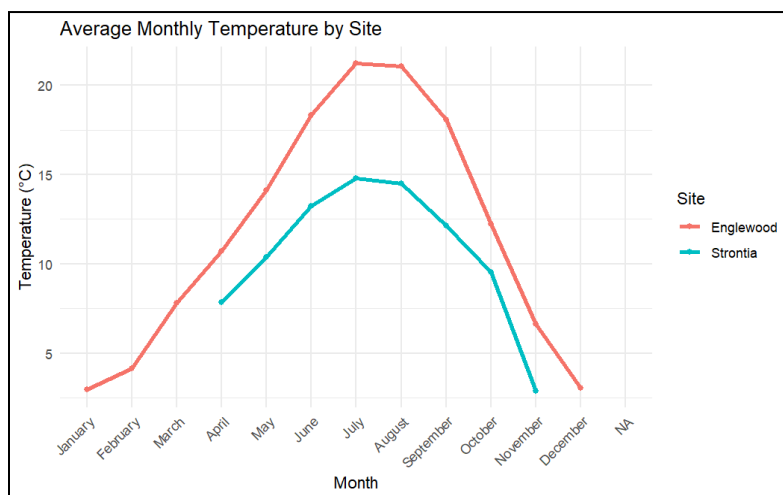


Figure 3: Plot showing average monthly water temperature for the Englewood and Strontia Springs sites in the South Platte Basin. Data used to calculate averages for Englewood ranged from 1983-2024, and data for Strontia Springs covers the years 2022-2024.

When looked at by month, Englewood demonstrates elevated temperatures throughout the year, with the most significant differences occurring during the summer months (June–August). These results align with our expectation that urban areas experience greater heating, especially during low-flow periods typical of late summer. The urban landscape absorbs and re-radiates heat, and reduced vegetation cover limits cooling through shading and evapotranspiration. In contrast, surrounded by forested uplands and reservoir influences, Strontia Springs maintains cooler and more stable temperatures, especially during peak thermal stress periods.

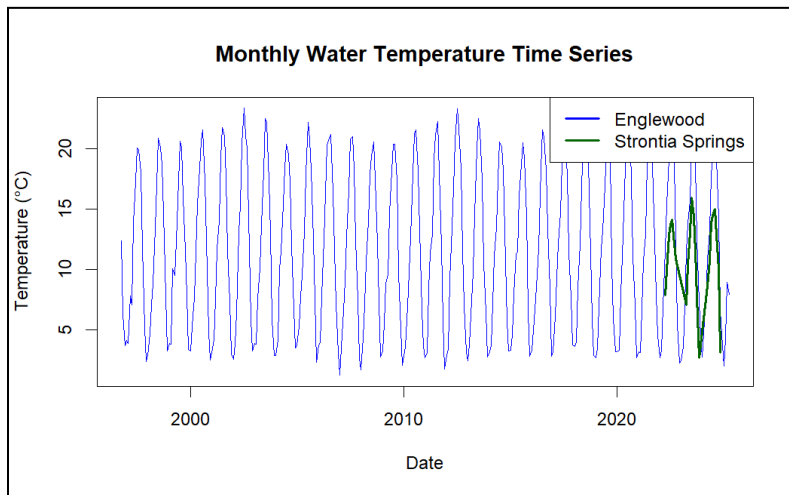


Figure 4: Time series of monthly water temperature for the Englewood and Strontia Springs sites in the South Platte Basin. Data for Englewood ranges from 1983-2024, and data for Strontia Springs covers the years 2022-2024.

The full-time series visualization illustrates that both sites follow a similar seasonal temperature cycle, with lows in winter and highs in summer. However, Englewood experiences more extreme high temperatures and quicker daily fluctuations, supporting our hypothesis that urban areas alter physical water characteristics. These spikes may be linked to stormwater runoff events or daily heating associated with tight surfaces and low canopy cover. The more stable temperature profile at Strontia Springs indicates stronger buffering capacity in upstream environments.

This temperature analysis supports our hypothesis that urbanization along the South Platte, particularly through Denver, results in lower water quality conditions, as evidenced by increased water temperatures. Elevated temperatures can reduce dissolved oxygen levels and stress aquatic ecosystems, making this a critical parameter for watershed management. The findings also support the need to examine other metrics like discharge, turbidity, and pollutant concentrations to understand how Denver’s urban footprint affects the river healthfully.

Discharge

For discharge analysis we wanted to evaluate 1) how the South Platte River changes from the headwaters, to Denver, and past Denver and 2) how the South Platte River compares to the North Platte River and the contribution to the Platte River.

In order to test our first question we compared discharge from the Hoosier Pass diversion point at the start of the South Platte in the Rocky Mountains, discharge from Englewood just before Denver, Commerce City just after Denver, and discharge from Fort Morgan in the Eastern plains of Colorado with observations starting from around 2000. Each of these locations have 5 variables including: agency code, site number, date, data (in cubic feet per second), and data code which was converted to acre feet per year. Discharge data was collected daily starting in January of 1983 for all sites except Hoosier Pass, which began in 1984. These sites were chosen based on location and the limited discharge data that was available along the river. Comparing all three sites over time and separated for Water Year (*Figure 5 and 6*) it is clear that as the South Platte passes through Denver, discharge has historically increased throughout the river. This is surprising and debunks our second hypothesis that Denver will negatively influence water availability along the South Platte River. In fact it continues to gain discharge even through the plains where agriculture is prevalent. Notable in these figures is also the 2013 flood that affected the Big Thompson River (City of Loveland n.d.). In order to avoid its bias we focused on years when Fort Morgan data was available and excluding the flood (*Figure 6*). The discharge values for Fort Morgan still exceed those of the South Platte before reaching this point.

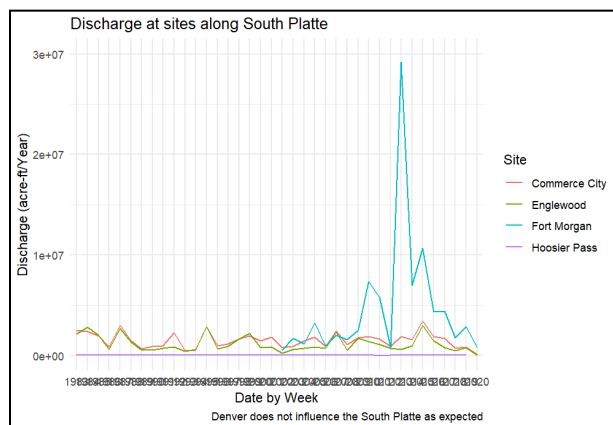


Figure 5: Discharge data at four sites along South Platte River from 1983-2024.

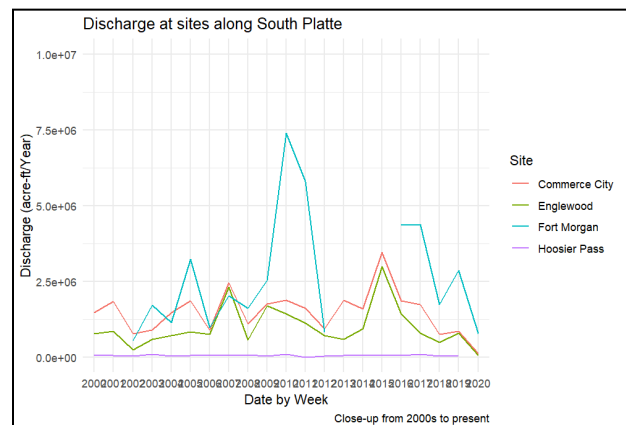


Figure 6: A close-up of discharge data at four sites along the South Platte River from 2000-2024.

Our second discharge related question involved comparing the North Platte River and the South Platte River, as well as their contribution to the Platte River. We evaluated the Commerce City, Colorado site as it had the most data furthest along the South Platte River with 13,515 observations, a location at the Wyoming/Nebraska border along the North Platte River as this was also the most available location along the river providing the most data, and the Platte River

at Overton, Nebraska for the same reasons discussed before. Through our analysis we found that the North Platte on average, contributes more water than the South Platte to the Platte River (*Figure 7*). Our second hypothesis is also falsified through this time series graph. The North Platte also reveals similar peaks to the Platte River, which indicates that it has more of an influence on the entire river system, when compared to the South Platte.

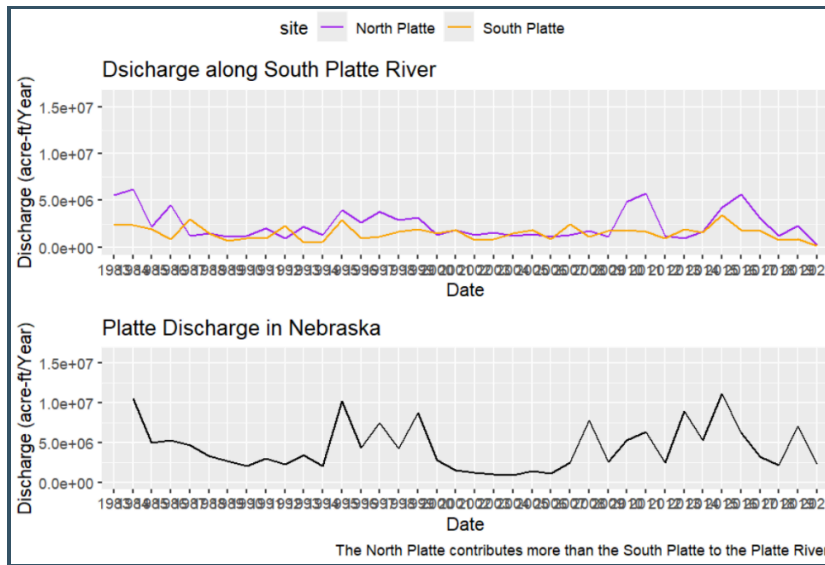


Figure 7: The top plot shows discharge levels along the North and South Platte River from 1983-2024, and the bottom plot shows Platte discharge data in Nebraska from 1983-2024.

4. Conclusion

Through our exploratory data analysis of the South Platte Basin, we gained valuable insights into how land use, urbanization, and water management practices shape both water quantity and quality along the river's course. Our discharge analysis showed that while flow generally increases downstream from the headwaters due to tributary contributions and major diversions such as Hoosier Pass, there is a small increase in flow through and just after Denver, which we attribute to wastewater treatment (Metro Water Recovery n.d.). This surprise increase in water reflects substantial allocation treatment of water resources, as illustrated by *Figure 7*. Due to the influence of return flows, particularly treated effluent released by the Denver Metro Wastewater Recovery Facility, which is one of the largest treatment plants in the region, the water quantity goes up.

Our analysis of water chemistry further revealed how urbanization and return flows impact water quality. As expected, we observed notable increases in water temperature downstream of Denver, suggesting elevated levels of dissolved solids and altered thermal regimes. In particular, our temperature data showed consistently higher annual average and average monthly water temperatures in Englewood, downstream of Denver, compared to lower temperatures recorded at Strontia Springs, which is located upstream of the urban corridor. This spatial temperature pattern further reflects the influence of urban heat inputs, reduced riparian

shading, and the thermal loading associated with wastewater effluent. These patterns are consistent with increased pollutant loads and anthropogenic inputs that accompany urban land cover and infrastructure. The results echo broader concerns about water quality degradation in the South Platte, particularly in stretches where water reuse, stormwater runoff, and effluent discharges are prevalent. These findings are especially significant given the South Platte's role as a primary water source for both urban populations and agriculture throughout northeastern Colorado and into Nebraska.

In conducting this analysis, we faced several data-related challenges, including gaps in temporal coverage and inconsistencies between chemistry and discharge datasets. These limitations highlighted the importance of transparent data documentation, careful site selection, and the need for integrated, long-term monitoring programs that support robust spatiotemporal analysis. The fragmented nature of publicly available data also revealed an opportunity for improving data infrastructure across watersheds, particularly those experiencing rapid urban expansion or increasing water stress.

Overall, our findings emphasize the need for expanded monitoring and proactive water management in the South Platte Basin. As climate change, population growth, and intensifying water demands continue to pressure the system, it is essential to understand not just natural hydrologic patterns but also the engineered and institutional drivers that influence water availability and quality. Return flows also play a major role in augmenting downstream discharge and shaping water chemistry. Recognizing the benefits and risks of relying on such flows is critical for future water planning, especially as water reuse and recycling become more prominent in arid and semi-arid regions.

Looking ahead, a valuable next step would be to conduct a more detailed temporal analysis, examining how discharge and water quality metrics vary across seasons, water years, and hydrologic events, like snowmelt or monsoon periods. Further, analyzing concentrations of specific pollutants, such as nitrate, phosphorus, or heavy metals, would provide a clearer picture of urban and agricultural impacts on aquatic ecosystems and human health. It would also be useful to explore how major diversions alter hydrologic connectivity and ecological function.

Finally, integrating our findings into a broader systems framework, accounting for institutional decision-making, legal constraints (interstate compacts and prior appropriation), and socio-economic drivers, would support the development of holistic, adaptive, and equitable water management strategies. The South Platte is not only a physical river but also a managed and contested resource, and addressing its challenges requires both scientific insight and stakeholder collaboration. Our exploratory analysis serves as a foundation for such interdisciplinary work and points to the need for sustained investment in data collection, synthesis, and inclusive water governance.

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